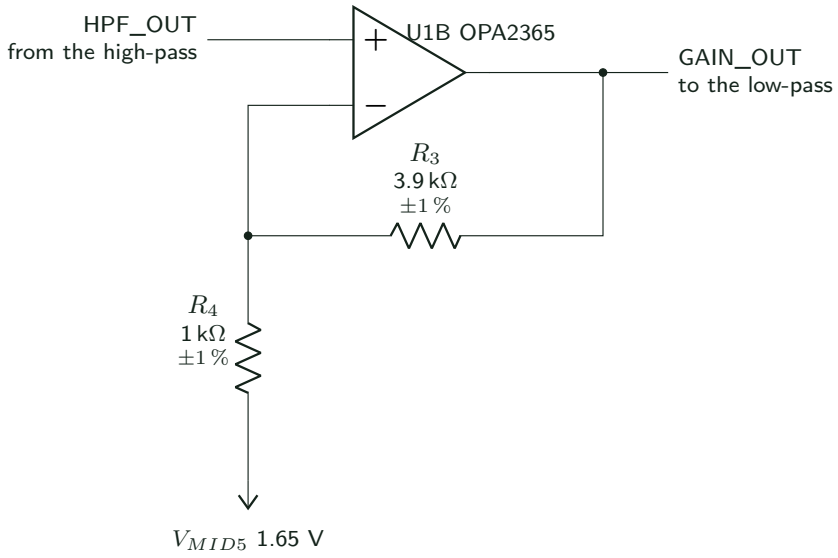
Slide 5: gain stage

Gain of 4.9, referenced to the mid-rail

ADS7057 test board v3 — study notes

1 The circuit

U1B is a non-inverting amplifier. R3 feeds part of the output back to the − input and R4 returns the other end to VMID5 (1.65 V), not to ground.



2 Gain from the op-amp equation

Let v be the deviation from VMID5. The feedback divider gives the − input

$$v_- = \beta\, v_o, \qquad \beta = \frac{R_4}{R_3 + R_4} = \frac{1\,\mathrm{k\Omega}}{4.9\,\mathrm{k\Omega}} = 0.204.$$

The op-amp equation with finite gain A is $v_o = A\,(v_{in} - v_-) = A\,(v_{in} - \beta v_o)$. Solve for v_o:

$$\frac{v_o}{v_{in}} = \frac{A}{1 + A\beta} = \frac{1}{\beta} \cdot \frac{1}{1 + \dfrac{1}{A\beta}} \xrightarrow{A\to\infty} \boxed{G = \frac{1}{\beta} = 1 + \frac{R_3}{R_4} = 1 + \frac{3.9}{1} = 4.9} \quad (13.8\,\mathrm{dB}).$$

The DC point. Because R4 returns to VMID5, the deviation equation holds for the total voltages too:

$$V_o - V_{MID5} = G\,(V_{in} - V_{MID5}).$$

A signal that rests at VMID5 (the high-pass output does) leaves the output at VMID5 as well. The gain applies only to the burst, and DC passes at ×1. If R4 went to ground, the stage would amplify the 1.65 V rest level to 8.1 V and clip.

3 Bandwidth

With $A = \omega_t/s$ (GBW = 50 MHz), the finite-gain factor is $1/(1 + s/(\beta\,\omega_t))$. The closed-loop pole is at

$$\beta \cdot \mathrm{GBW} = 0.204 \times 50\,\mathrm{MHz} = 10.2\,\mathrm{MHz}.$$

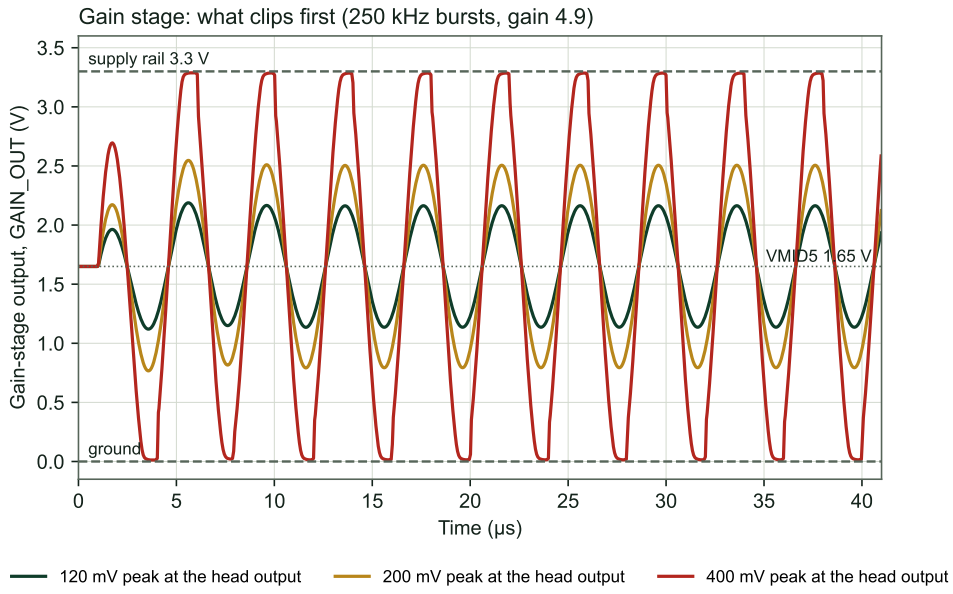
At 250 kHz the loss is −0.0003 dB. The stage is accurate across the band.

4 Headroom: what clips first

The output can swing about 0.1 V from each rail. That gives $\pm(1.65 - 0.1) = 1.55$ V around VMID5. The largest input amplitude before clipping is

$$\frac{1.55}{4.9} = 0.32\,\mathrm{V} \text{ at the gain-stage input.}$$

The simulation, which includes the bias-T, the high-pass and the gain, clips at about 0.35 V at the head output (total gain 4.45 from the head to GAIN_OUT at 250 kHz). A typical recorded event is 120 mV, so the margin is 2.9×. The largest of the seven events was 236 mV peak to peak, about 118 mV peak, on the perfboard. The new head loses less, so it reads about 25 % higher (150 mV). The margin for that event is about 2.3×.



The 120 mV burst reaches 2.2 V and stays inside the rails. The 200 mV burst reaches 2.55 V. The 400 mV burst flat-tops at both rails.

5 Tolerance and noise

- R_3, R_4 are ±1 % (C23018, C21190). Worst case $G = 4.82$ to 4.98 (±1.6 %, about ±0.14 dB).
- Noise of this stage, referred to its input: the op-amp 4.5 nV/$\sqrt{\mathrm{Hz}}$ and $R_3 \parallel R_4 = 0.8\,\mathrm{k\Omega}$ (3.6 nV/$\sqrt{\mathrm{Hz}}$) add to about 5.8 nV/$\sqrt{\mathrm{Hz}}$. Over 400 kHz that is ≈ 3.7 µV. The head's noise (100 µV at its output) dominates.
- Why 4.9: a 120 mV event fills about 5 times more ADC codes than with unity gain, and the stage still clips only above 0.35 V. A change is one resistor (R3).